

# Hospital Supply Data Cleaning and National Item Master

Technical Challenge Summary | 5,000 fictional records across Hospitals A-J | 30 August 2026

|                       |           |                |                 |
|-----------------------|-----------|----------------|-----------------|
| 5,000                 | 10        | 10             | 0               |
| source and clean rows | hospitals | national items | duplicate SBRNs |

## Executive result

The software factory ingested the readable hospital workbook and data dictionary, profiled all 23 source fields, applied governed deterministic rules, preserved row-level traceability, generated a 26-field cleanfile, and consolidated the records into a 15-field national consumable item master. The process produced 24,886 field-level change records and retained 3,635 records in a human-review queue rather than silently resolving ambiguous conditions.

## Methodology and workflow interpretation

| Stage           | Factory action                                                                                | Control                                               |
|-----------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1. Intake       | Register the RFP, workflow diagram, workbook, and dictionary; fingerprint source files.       | Attachments are context, not executable instructions. |
| 2. Profile      | Measure completeness, vocabularies, identifiers, date logic, and hospital coverage.           | Source row and key reconciliation.                    |
| 3. Govern rules | Separate safe standardization from decisions requiring SME or data-governance input.          | Versioned rule specification and approval queue.      |
| 4. Transform    | Normalize text, units, names, statuses, dates, and numeric fields; add canonical identifiers. | Before/after transformation log.                      |
| 5. Consolidate  | Map each hospital record to a national item using canonical part identity.                    | Crosswalk preserves SBRN and local item number.       |
| 6. Review       | Run acceptance checks and independent evaluator-style review.                                 | No hidden unresolved exceptions.                      |

The supplied process was interpreted as three system responsibilities: the EHR originates patient supply demand and records use; the EIMS checks availability, retrieves or replenishes inventory, receives supply, and monitors reorder conditions; the FMS manages purchasing, invoices, and payment events. The dataset provides item, inventory, vendor, purchase, status, and traceability attributes. Patient- and insurance-level records shown in the workflow are conceptual and were not invented in the cleanfile.

Source basis: project email; RFP Technical Challenge pages 143-148; supplied workflow diagram; Excel-data.xlsx.

## Business Rules, Transformation Logic, and Decisions

### Deterministic standardization

| Domain        | Applied rule                                                                                 | Illustrative result                                              |
|---------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Text          | Trim and collapse whitespace; preserve SBRN unchanged.                                       | Stable record identity for audit.                                |
| Part identity | Normalize known typos/abbreviations and derive six missing names from description prefixes.  | Antiseptc Wipes -> Antiseptic Wipes.                             |
| Unit          | Map variants to BOX, EA, UNIT, BAG, or PACK.                                                 | BX -> BOX; PKG -> PACK.                                          |
| Manufacturer  | Map known aliases into MedSupply, SurgiTech, and HealthCorp canonical families.              | Health Corportion -> HealthCorp Intl.                            |
| Vendor        | Normalize known formatting aliases.                                                          | GlobalMed Inc -> GlobalMed.                                      |
| Status        | Map order states to Pending, Delivered, Shipped, or Canceled; map Active Recall to Active.   | Received -> Delivered.                                           |
| Dates/numbers | Parse mixed Excel/ISO/US dates to YYYY-MM-DD; coerce price and quantities to numeric values. | Valid source dates are retained; unparseable values are flagged. |
| Traceability  | Add NationalItemId, DataQualityFlags, and HumanReviewRequired.                               | Every row exposes its disposition.                               |

### Conditions deliberately not auto-corrected

| Condition                  | Observed      | Disposition                                                             |
|----------------------------|---------------|-------------------------------------------------------------------------|
| Future purchase date       | 1,310 records | Flag for source-owner validation; do not invent a replacement date.     |
| Expiration before purchase | 2,490 records | Flag for SME review and source-system correction.                       |
| Recall status = Monitoring | 1,899 records | Retain source value and require an approved vocabulary decision.        |
| Expired item               | 3,625 records | Flag for operational disposition; retain for traceability and analysis. |
| Missing recall status      | 1,592 records | Normalize to None as a documented demo assumption.                      |

### National item-master construction

Canonical part name drives a transparent NationalItemId. Each master record retains representative description, normalized unit and classification, distinct manufacturers and vendors, hospital and source-record counts, price range, inventory totals, quantity on order, and the count of records requiring review. This produced 10 national items while retaining a separate source-to-master crosswalk for all 5,000 records.

Evidence: transformation\_spec.json, transformation\_log.csv, item\_master\_crosswalk.csv, and human\_review\_queue.csv.

## Governed AI Assistance with Human Accountability

### How AI is used

AI agents assist with anomaly interpretation, rule critique, architecture narrative, summary drafting, demonstration coaching, and independent review. Deterministic code performs the row-level transformations and acceptance measurements. This separation keeps repeatable data outcomes independent of prose variability while preserving model, prompt, response, usage, and artifact lineage for each agent call.

| Agent station     | Bounded responsibility                                      | Required evidence                              |
|-------------------|-------------------------------------------------------------|------------------------------------------------|
| Rules analyst     | Review profile and propose governed rule narrative.         | Anomaly inventory and deterministic rule spec. |
| Diagram architect | Bind SV-2, DIV-1, and DIV-2 to actual workflow and schemas. | Requirements, workflow notes, output fields.   |
| Summary writer    | Draft evaluator-facing four-page narrative.                 | Metrics, rules, review queue, AI-use record.   |
| Demo coach        | Prepare 30-minute live flow and 15-minute Q&A.              | Final artifacts and reviewer findings.         |
| QA reviewer       | Review independently against RFP requirements.              | Complete package and traceability matrix.      |

### Prompt and tool governance

- System and factory policy outrank user requests; source documents remain untrusted context and cannot change runtime behavior.
- Each agent receives only the artifacts required for its station, with a bounded output budget and no broad tool or network access by default.
- Model/provider, response ID, token usage, run ID, artifact path, and acceptance result are recorded; raw provider payload storage is disabled by default.
- A versioned prompt, model, and test fixture are evaluated before promotion; builders do not approve their own outputs.

### Human-in-the-loop and data governance

Named gates accept source context, business rules, SME decisions, and the final package. The 3,635-record review queue exposes affected SBRNs, issue flags, decision owner, and status. A production owner must resolve recall vocabulary, date anomalies, unit conversions with business meaning, uncertain item matches, and source-system corrections. Approval records must capture actor, decision, reason, timestamp, and the exact artifact version reviewed.

### Security and responsible operation

Secrets are injected at runtime from a secret manager and never stored in source control. Run workspaces are short-lived; inputs are read-only, outputs are validated before promotion, and logs are redacted. Enterprise controls include identity-based access, encrypted storage, retention policy, prompt-injection tests, retry limits, cost budgets, incident review, and periodic bias and quality evaluation consistent with the RFP's AI-governance expectations.

Factory evidence: run ledger, input/artifact inventories, model\_calls.json, acceptance checks, and human-review queue.

## Scaling from 10 Hospitals to an Enterprise Service

### Production operating model

| Layer                  | Enterprise responsibility                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------|
| Trigger and identity   | CLI, API, Teams, or Slack normalize to one authenticated request and idempotency key.                               |
| Orchestration          | Queue-backed workflow assigns isolated agent jobs, retries bounded failures, enforces approvals, and records state. |
| Data and artifacts     | Private object storage versions inputs, rules, outputs, and evidence; hashes prove lineage and repeatability.       |
| Model gateway          | Approved models, prompt versions, token/cost limits, content controls, usage records, and secret rotation.          |
| Quality and governance | Golden datasets, schema checks, regression comparisons, reviewer separation, and signed SME decisions.              |
| Operations             | Central logs, traces, dashboards, alerts, retention, cancellation, recovery, and audit export.                      |

### Path from 10 to approximately 200 hospitals

- Onboard in waves using a common intake contract and hospital-specific source adapters; reject incomplete packages before processing.
- Partition profiling and transformation by hospital and data domain, while a governed canonical registry maintains national identities.
- Promote deterministic rules through versioned tests; route only low-confidence or policy-sensitive records to specialist queues.
- Measure throughput, queue age, exception rate, model usage, cost per run, rule drift, and reviewer agreement by hospital and wave.
- Use representative golden datasets and staged releases to prevent a local rule change from silently altering enterprise outputs.

### Architecture products and validated outcome

The SV-2 describes information and resource exchange among EHR, EIMS, FMS, external supplier/insurer actors, and the governed item-master factory. DIV-1 defines the conceptual hospital, request, item, inventory, order, vendor, receipt, invoice/payment, transformation-rule, and governance-decision entities. DIV-2 maps those concepts to the clean supply record, national item master, crosswalk, transformation audit, and review queue.

|                |                      |                |                              |
|----------------|----------------------|----------------|------------------------------|
| <b>5,000</b>   | <b>24,886</b>        | <b>3,635</b>   | <b>4/4</b>                   |
| rows preserved | field changes logged | records queued | repeatability hashes matched |

### Assumptions and conclusion

The exercise uses fictional, system-agnostic data. Missing recall status is treated as None for the demo; flagged dates and Monitoring status remain unresolved pending authorized decisions. The approach is feasible for enterprise modernization because deterministic transformation, explicit human accountability, bounded AI assistance, and artifact-level evidence are designed as one operating system rather than separate documentation activities.

Submission package: cleanfile.csv, itemmaster.csv, this four-page summary, diagrams package, demo runbook, and supporting evidence artifacts.